

Polymarket Copy Trader

Automatically mirror trades of top-ranked wallets on Polymarket

1. Setup (one-time)

Install dependencies and configure:

```
pip install -r requirements_copy_trader.txt
cp .env.example .env
nano .env      # fill in your wallet private key + settings
```

Run:

```
# Paper trade (safe, no real money) – default
python copy_trader.py

# Live trading
PAPER_TRADE=false python copy_trader.py

# Crypto markets (Bitcoin)
MODE=btc python copy_trader.py
```

2. How It Works

Step	What happens
1. Load wallets	Reads top multi-period wallets from your data file
2. Poll subgraph	Every 15 s checks for new trades by those wallets on tracked markets
3. BUY signal	When MIN_COPY_WALLETS (default 2) agree → places a GTC limit order
4. SELL signal	When those wallets start exiting → mirrors the exit
5. Sizing	25% fractional Kelly criterion, capped at MAX_POSITION_USDC

3. Key .env Settings

Variable	Default	Meaning
POLYMARKET_PRIVATE_KEY	—	Your wallet's private key (Polygon mainnet)
PAPER_TRADE	true	Set false for real orders
COPY_BUDGET_USDC	50	Total USDC pool for all copy positions
MAX_POSITION_USDC	20	Cap per single market
MIN_COPY_WALLETS	2	Wallets that must agree before copying
MODE	war	war, btc, eth, or sol
POLL_INTERVAL	15	Seconds between subgraph polls
MAX_YES_PRICE	0.90	Don't buy if YES > 90 cents (overpriced)
MIN_YES_PRICE	0.03	Don't buy if YES < 3 cents (too risky)
SUCCESS_RATE	0.82	Assumed win-rate used for Kelly sizing

■ **BEFORE** setting `PAPER_TRADE=false`: • Fund your wallet with USDC on Polygon mainnet at app.polymarket.com • Test in paper mode first and verify signals look correct • Never commit your .env file to git (it contains your private key)

4. Copy-Trading Signal Logic

A BUY order fires when:

- At least MIN_COPY_WALLETS tracked wallets have bought the same market
- Current YES price is between MIN_YES_PRICE and MAX_YES_PRICE
- You do not already hold a position in that market
- Kelly sizing gives an allocation of at least \$0.50

A SELL order fires when:

- At least MIN_COPY_WALLETS tracked wallets sell / reduce their position
- You currently hold tokens in that market

Position sizing (25% fractional Kelly):

```
b = (1 / price) - 1 # net odds on a win
Kelly = (b * SUCCESS_RATE - (1-SUCCESS_RATE)) / b
allocation = min(BUDGET * Kelly * 0.25, MAX_POSITION_USDC)
```

5. Which Wallets Are Tracked?

The bot reads the top wallets from your local data files, generated by GitHub Actions every 2 hours (or on demand):

MODE	Data file	Wallet source
war	data/war_data.json	mp_wallets (profitable in Feb...26 AND Mar...26)
btc/eth/sol	data/crypto_data.json	per-coin mp_wallets + top50

Config reloads every ~3 minutes during runtime to pick up newly discovered wallets.

6. Files

File	Purpose
copy_trader.py	Main bot script
requirements_copy_trader.txt	pip dependencies (requests, py-clob-client, python-dotenv)
.env.example	Config template — copy to .env and fill in values
copy_trader.log	Auto-created log of all signals and orders